

A Low-Power Variation-Tolerant 7T SRAM With Enhanced Read Sensing Margin for Voltage Scaling

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Abstract—Reducing the minimum operating voltage ($V_{\min}$) and improving the variation tolerance are the main design challenges of voltage-scalable static random access memories (SRAMs). This article presents a variation-tolerant 7T SRAM with reduced data-dependent read bitline (RBL) leakage to enhance read sensing margin and improve $V_{\min}$ without any assist techniques. The design implements a delay-tracking-based adaptive timing generation technique and an error detection circuit for process, voltage and temperature variation-tolerant operation and reliable dynamic voltage scaling. An 8-kb SRAM macro is implemented in 55-nm CMOS technology to demonstrate our design. Post-layout simulation results show error-free full functionality down to 0.35 V at an operating frequency of 202 kHz. The RBL leakage current for 7T SRAM cell is improved by 2.6 $\times$ and the $I_{\text{on-to-}I_{\text{off}}}$ ratio is improved by 2.3 $\times$ compared to the conventional 8T SRAM cell. The minimum energy of 1.76 pJ is achieved at 0.4 V. The proposed 7T SRAM functioning as an on-chip memory in a voltage-scalable system on chip is measured and demonstrate operation from 0.75 to 1.2 V at 32 MHz. The average power consumption for the entire chip performing 7T read and write operations decreased by 19.3% compared to that of the standard 6T SRAM at 0.75 V.

Index Terms—Dynamic voltage scaling, low power, low voltage, read margin, static random access memory (SRAM).

I. INTRODUCTION

WITH the growing market of power-sensitive products, such as portable devices, biochips, and medical electronics, dynamic voltage frequency scaling (DVFS) is commonly used to obtain both higher performance and lower energy consumption. Since supply voltage and frequency are two key factors that determine active power dissipation, DVFS is able to dynamically adjust the operating frequency and supply voltage of the processor to match the performance requirements in different application scenarios. For example,

the supply voltage and frequency are increased to enable high-speed operation when the processor is running at high throughput. In standby mode, the frequency and supply voltage are lowered to reduce power dissipation. The DVFS technique allows the system to operate at an appropriate voltage and frequency with full functionality, thereby maximizing energy efficiency and extending battery life.

Embedded static random access memory (SRAM) is the most widely used memory for on-chip data storage and consumes significant fraction of power and area [1]. Thus, it is imperative for DVFS designs to have SRAMs capable of operating over a wide supply voltage range. The conventional 6T SRAM fails to operate in the subthreshold region due to the read disturb and half-select issues. Many new bit-cell topologies and assist techniques have been reported to enhance SRAM reliability at low voltages [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. However, while some bit-cell topologies improve the low voltage read (write) operation by weakening the positive feedback [2], separating the read and write ports [3], [4], [5], [6], and improving bitline sensing margins [8], [9], [10], [11], they also introduce hold stability and density loss. Some designs require additional assist techniques to ensure high-speed access at low voltage due to their stacked read (write) ports [7]. An assist technique is often beneficial for a specific restricted operation and useless or even detrimental for other operations. It can hardly be used to improve SRAM performance over a wide voltage range. In addition, the power dissipation of the assist techniques diminishes the benefits of lower supply voltage.

In this article, we report a variation-tolerant 7T SRAM with reduced data-dependent read bitline (RBL) leakage to enhance read sensing margin and improve the minimum operating voltage. An adaptive timing generation circuit is employed to track process, voltage, and temperature (PVT) variations and ensure robust operation over a wide voltage range. The main contributions of this article are: 1) proposing a 7T SRAM cell with reduced data-dependent RBL leakage to improve $V_{\min}$; 2) providing tolerance to variation and extending the operating voltage range of the memory by employing a delay-tracking-based adaptive timing generation (DTBATG) circuit; and 3) adding a lightweight error detection circuit based on DTBATG to indicate the boundary of voltage scaling. The combination of above techniques enables low-power, robust

Manuscript received 20 September 2023; revised 5 December 2023; accepted 12 February 2024. Date of publication 19 February 2024; date of current version 19 July 2024. This work was supported by the National Natural Science Foundation of China under Grant 62274068. This article was recommended by Associate Editor M. Mutyam. (*Corresponding author: Zhenglin Liu.*)

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Digital Object Identifier 10.1109/TCAD.2024.3367233

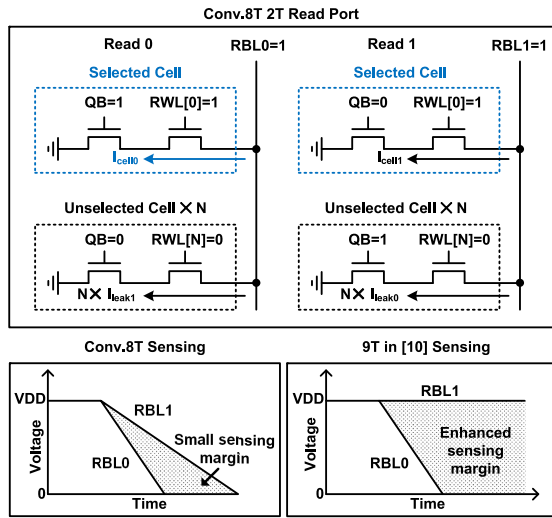


Fig. 1. RBL sensing margin of conventional 8T SRAM cell.

SRAM operation over a wide voltage range without any assist techniques.

The remainder of this article is organized as follows. Section II reviews the challenges of designing voltage-scalable SRAMs and presents some state-of-the-art similar SRAM designs. Section III describes the proposed 7T SRAM cell and its basic operations, including RBL leakage analysis and low-power read operations. This is followed by the proposed timing optimization technique DTBATG and an error detection circuit based on DTBATG in Section IV. The post-layout simulation and test chip measurement results of the SRAM macro based on the 55-nm CMOS process and the discussions with recent works are detailed in Section V. Section VI presents the conclusion.

II. CHALLENGES OF SRAM VOLTAGE SCALING

Voltage-scalable SRAM designs face the challenges of lowering the minimum operating voltage and optimizing the performance of SRAMs over a wide voltage range. As the active power dissipation increases quadratically with the supply voltage, extending the supply voltage to the subthreshold region has a positive impact on energy efficiency. The main challenges preventing SRAMs from operating at very low voltages are the significant variation threshold voltage (V_t) of transistors caused by random doping fluctuation (RDF) and severely degraded I_{on} -to- I_{off} ratio in the subthreshold region [23]. Although SRAMs with decoupled read ports (RPs) are capable of operating in the subthreshold region [3], [4], [5], [6], they still suffer from diminished bitline voltage swing caused by the degraded I_{on} -to- I_{off} ratio.

Fig. 1 illustrates the 2T RP of the conventional decoupled 8T SRAM. For a bitline with $N + 1$ bit-cells, there are N leakage current paths during a read operation is performed. In the subthreshold region, the decayed read current is comparable to the total leakage current, making it more difficult to sense bitlines for reading “0” and “1.” In addition, the leakage of the 2T RP is data-dependent. The worst case of the overall pull-down currents for reading “0” and “1” are

“ $I_{cell0} + NI_{leak1}$ ” and “ $I_{cell1} + NI_{leak0}$,” respectively. Given that $I_{leak0} > I_{leak1}$, the bitline discharge for read “0” may be faster than that for read “1,” leading to incorrect sensing. Some RP decoupled SRAM cells have been reported [8], [9], [10], [11] to enhance bitline sensing margins by optimizing bitline leakage. However, the 9T SRAM in [8] retains the worst-case leakage current I_{leak0} and requires a current bias to compensate for bitline leakage. The 10T SRAM cell in [9] improves read performance by reducing data-dependent bitline leakage. However, it suffers from severely degraded read delay due to the triple-nMOS stacked RP. Another 10T SRAM presented in [10] does not experience degradation in read “0” discharging, but it introduces an internal leakage path when the bit-cell stores “0.” Additionally, the substantial area overhead of 10T SRAMs diminishes their appeal in on-chip memories.

The main limitation of new bit-cell topologies for wide voltage operation is that only one of the hold, read, and write operations is improved [12], while the minimum operating voltage of SRAM is determined by the maximum value among read – V_{min} , write – V_{min} and hold – V_{min} [8]. Assist techniques are often implemented in SRAMs to improve specific operations at low voltages [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. For instance, negative bitline (NBL) [15], [21] and transient voltage collapse (TVC) [17] are adopted to enhance the writability of SRAMs by reducing the competition between pass gate (PG) and pull-up (PU) transistors temporarily. However, TVC degrades the holding stability of unselected cells. Other assist techniques include dual-rail power supply [13], VDD boost [16], word-line underdrive (WLUD) [20], word-line overdrive (WLOD) [18], and negative GND [19]. However, dedicated circuits such as charge pumps and voltage converters are needed to implement SRAM assist techniques, resulting in additional area and power overhead. A recent work [22] combined the write assist (NBL) and read assist (Suppressed bitline, SBL), and proposed a charge-recycling circuit to reduce the active power dissipation of the assist techniques without incurring significant area overhead.

Decoupled SRAM cells eliminates the size constraints conflict of transistors that exists in the conventional 6T SRAM. Hence, the read (write) ability of the decoupled SRAM cells can also be improved by adjusting the size ratio of the transistors. In comparison with assist techniques, sizing solutions are much easier to implement. However, the beneficial effects obtained by adjusting the ratio between transistor widths may be offset because of the increasing mismatch between transistors in the subthreshold region. In addition, the reverse short-channel effect (RSCE) becomes more pronounced due to the exponential dependence of drive current on threshold voltage and reduced drain-induced barrier lowering (DIBL) effect, resulting in a decline in V_t with longer channel length [24], [25]. Compared to designs with minimum channel length, longer channel lengths can achieve better performance in the subthreshold region and increase the robustness of the transistors to variations [26]. Fig. 2 depicts the effect of transistor length (L) on the read current of a 2T RP and static noise margin (SNM) of the 6T SRAM cell.

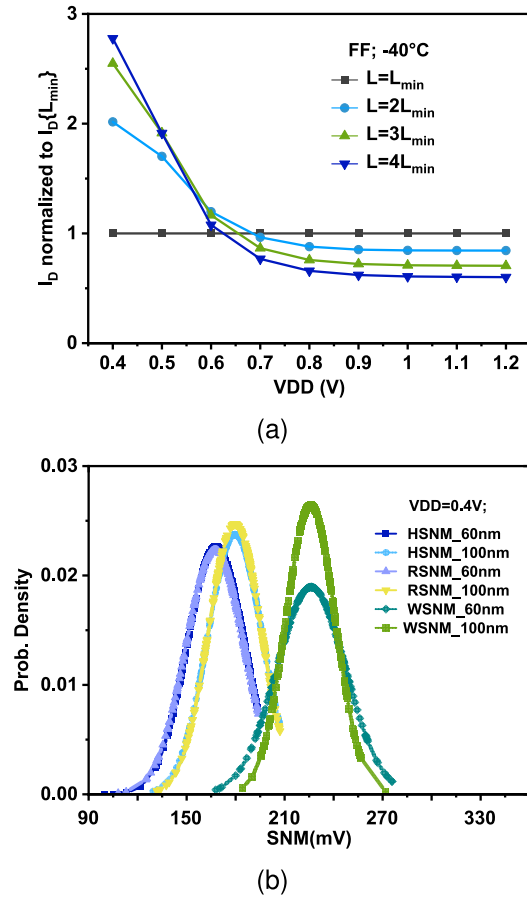


Fig. 2. Effect of transistor length on (a) read current of the 2T RP. (b) SNM of the 6T SRAM cell.

TABLE I
SNM DISTRIBUTION OF STANDARD 6T AND PROPOSED 7T

VDD	Cell	HSNM/mV		WSNM/mV		RSNM/mV	
		μ	σ	μ	σ	μ	σ
0.5V	6T	182.2	15.3	227.9	13.3	81.5	14.6
	7T	179.4	17.7	224.7	15.2	179.3	18.6
1.2V	6T	442.5	16.4	522.9	18.3	206.5	15.9
	7T	443.1	17.9	536.5	20.5	443.1	20.5

500 points were sampled using Monte Carlo (MC) simulation. The results shows that longer L causes a slight reduction in read current at high voltages. However, there is a significant increase in read current because of the RSCE at low voltages. Fig. 2(b) illustrates SNM distribution of the 6T SRAM cell at 0.4 V. Longer L improves the stability of read, write and hold operations simultaneously. The mean of hold SNM (HSNM) and read SNM (RSNM) distributions increased by 6.9% and 7.3%, while their standard deviations decreased by 4.6% and 9.6%, respectively. The standard deviation of write SNM (WSNM) decreased by 28%. Consequently, optimizing the sizing design emerges as a fundamentally cost-effective approach for reducing the SRAM operating voltage.

III. PROPOSED 7T SRAM DESIGN

The proposed 7T SRAM cell is illustrates in Fig. 3. It consists of a 1T RP and a 6T bit-cell topology for write and

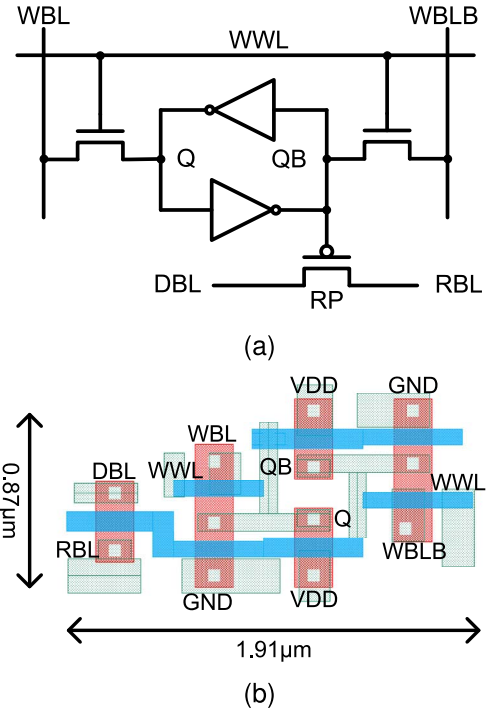


Fig. 3. Proposed 7T SRAM cell. (a) Schematic. (b) Layout.

hold operations. Data is stored on a pair of nodes of cross-coupled inverters, and write operations are performed through the write bitlines (WBL and WBLB) and write word line (WWL). The 7T SRAM performs the same write operation as the conventional 6T SRAM cell. The single-port read operation is performed through RBL and node DBL. The channel length of all the 7T transistors is set to 80 nm instead of the minimum length (60 nm) in 55-nm CMOS process to enhance SRAM performance at low voltages. We compared the performance and power consumption of the proposed 7T cell with the standard 6T cell at different voltages. Table I presents the SNM distributions of the two cell types obtained from 500-point MC simulation at 0.5 and 1.2 V. At 0.5 V, the RSNM distribution of the standard 6T cell no longer meets the 6σ yield requirement. As shown in Fig. 4, due to the decoupled RP, the mean of the proposed 7T cell RSNM distribution increased by 121% at 0.5 V and 115% at 1.2 V. Both cell types exhibit similar WSNM. The delay and power consumption for read and write operations are depicted in Fig. 5. The test circuits exhibit identical structures, with 64 cells connected to one single bitline. We tested the average power consumption and delay for write “1” and write “0” operations, and the average power consumption for read “0” and read “1” operations. For fair comparison, we define the read “1” delay as 20%*VDD differential voltage cross bitline pairs in the standard 6T cell and 60%*VDD single bitline voltage swing in the proposed 7T cell. The standard 6T cell and the proposed 7T cell demonstrate similar write power consumption and write delay. The read power consumption of the 7T cell is only 26% and 70% of that of the 6T cell at 0.7 and 1.2 V, respectively. However, the average read “1” delay for the 7T cell is $2.5\times$ of the 6T cell. This is

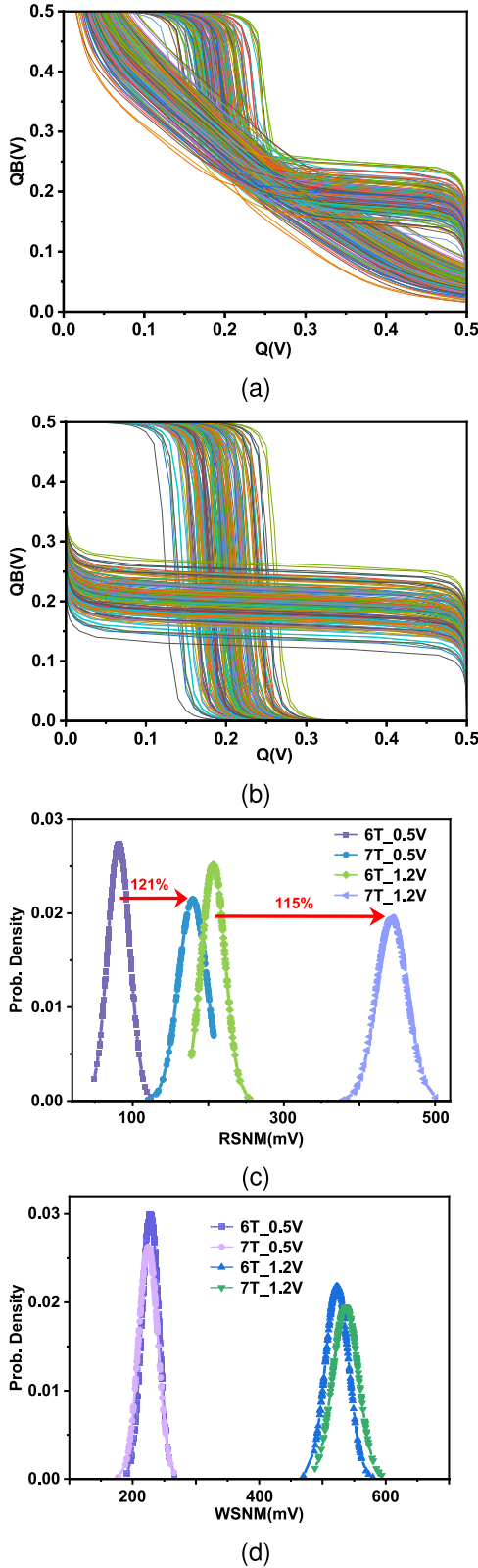


Fig. 4. SNM comparison of standard 6T and proposed 7T. (a) RSNM of standard 6T at 0.5 V. (b) RSNM of proposed 7T at 0.5 V. (c) RSNM comparison of standard 6T and proposed 7T. (d) WSNM comparison of standard 6T and proposed 7T.

attributed to the 7T cell adopting a single-ended large-signal read scheme, while the 6T cell employs a differential read scheme.

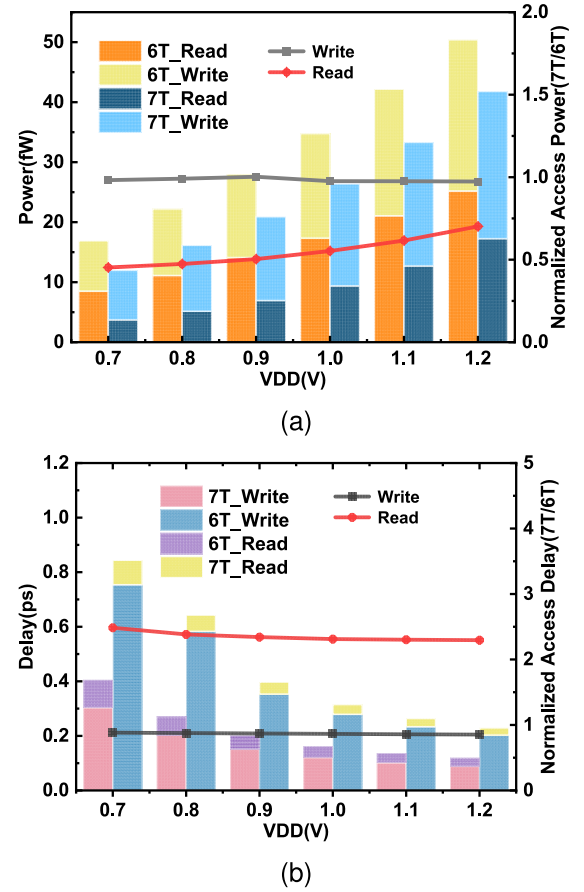


Fig. 5. Power and delay comparison of standard 6T and proposed 7T. (a) Power. (b) Delay.

A. Low-Power Read Operation Without Precharge

The read operation of the conventional 8T cell usually begins with RBL being precharged to “1,” and then RBL is selectively discharged to “0” or remaining in “1” depending on the data stored in the selected cell. However, RBL of the proposed 7T cell does not need to be precharged. Instead, RBL is discharged to “0” before performing the read operation. Fig. 6 shows the proposed 7T SRAM cell and peripheral circuits in a 128×64 SRAM macro. Node DBL is attached to the drain of a charging pMOS transistor and is shared by multiple SRAM cells in the same row. When the proposed 7T SRAM performing a read operation, RBL is first discharged to “0” through the pull-down nMOS transistor controlled by a discharge control signal (DCS), and then read word line (RWL) is asserted after DCS is negated. Node DBL attached to the selected cell is charged to “1.” If the selected cell stores data “1” (QB=0), RP is on and the PU current injected by VDD charges RBL to “1.” If the selected cell stores data “0” (QB=1), RP is off and RBL remains in “0.” However, to drive the pull-down nMOS transistor one extra signal DCS is to be routed for RBLs refreshing. The column-shared pull-down nMOS transistor is always turned on during the hold operation, and row-shared control signal DCS is negated and all RBLs in the array are floating during the read operation. In addition, for a one-bit column address, the RWL signal shared by each

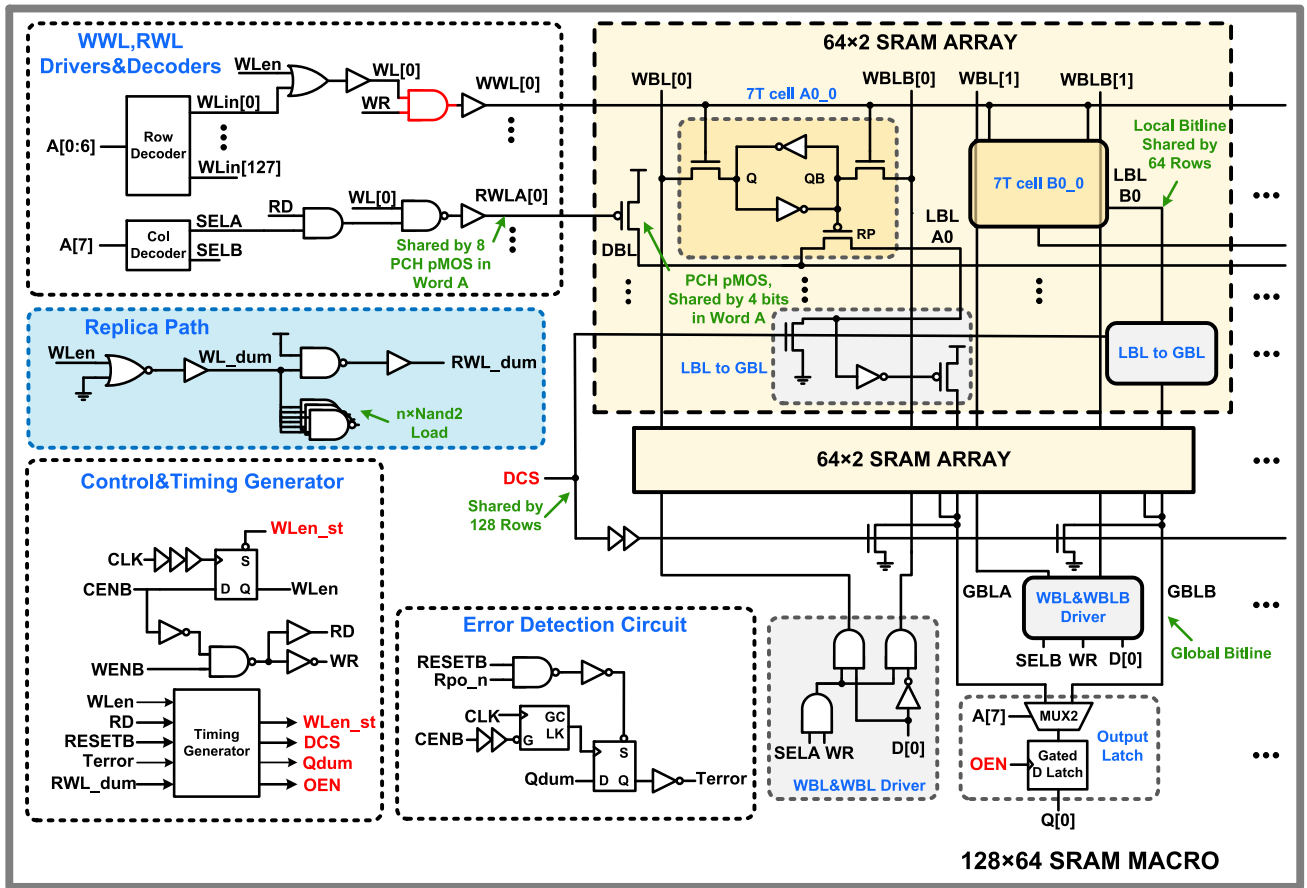


Fig. 6. Proposed 128×64 SRAM macro and the proposed 7T SRAM cell.

row is divided into two subsignals, $\overline{RWLA}$ and $\overline{RWLB}$. So that SRAM cells in the unselected columns will not consume active power during the read operation.

Single-ended large-signal sensing scheme is adopted to enhance the read margin at low voltages. As the read dynamic power consumption is directly proportional to the bitline capacitance (C_{BL}) and voltage swing (V_{swing}), optimizing the design to reduce both of these values is an effective approach to enhance energy efficiency. In our proposed 7T SRAM, we achieve this by implementing a hierarchical bitline strategy to reduce individual bitline capacitance and expedite sensing. As depicted in Fig. 6, the bitline in each column consists of a global bitline GBL and two local bitlines: LBL0 and LBL1. The latter is accomplished by negating RWL at the appropriate time, where RBL has not yet reached VDD (for the read “1” operation). Fig. 7 illustrates the waveform diagram of write “1” and read “1” operation. When GBL reaches “1,” RWL is immediately negated and stops charging RBL. At this time, the voltage of RBL V_{swing_7T} is lower than VDD. After the data is read out on the falling edge of OEN, DCS is asserted to refresh all bit lines to “0.” The RWL, OEN and DCS pulses are controlled and generated by the timing generator presented in Section III below. Compared to traditional 8T cell, achieving a reduced bitline swing by timing control during read “1” operation in 7T cell is more feasible. This is because the bitline PU current generated by the stacked pMOS is lower than the pull-down current generated by the stacked nMOS. During

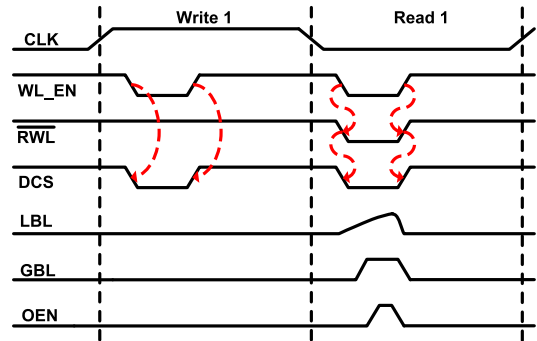


Fig. 7. Timing diagram of consecutive read and write operation.

read “0” operation, the power consumption in 8T cell primarily stems from RBL leakage, whereas in 7T cell, it originates from charging the shared node DBL and leakage accumulation in RBL. Since RP is off during read “0,” the charge on node DBL continues to be retained during RBL refreshing, which can be recycled in the next read period. Additionally, owing to the lower leakage current in 7T cell compared to 8T cells, the proposed 7T cell exhibit lower read power consumption in general.

The worst read operations occurs when all QBs in the same RBL store “0.” When charging the RBL, a portion of the current flows to the DBLs of other unselected cells in the same RBL, charging them as well. The corresponding read

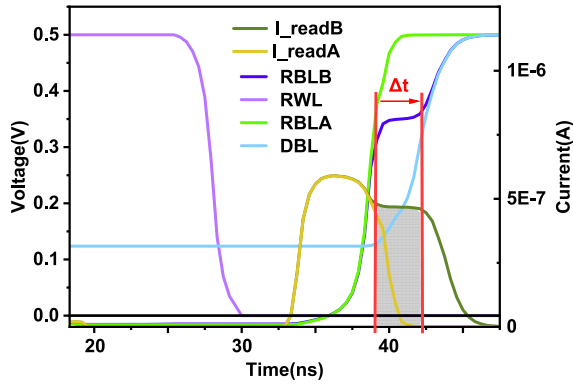


Fig. 8. Worst case read operation waveforms.

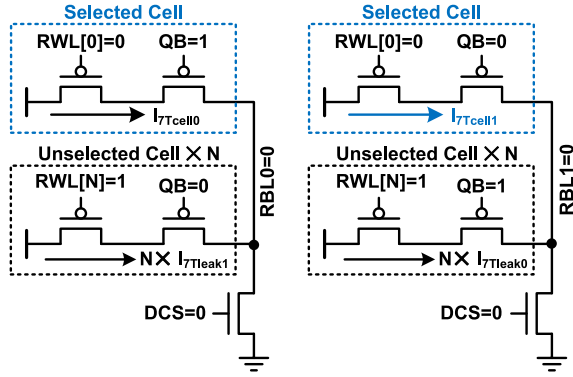
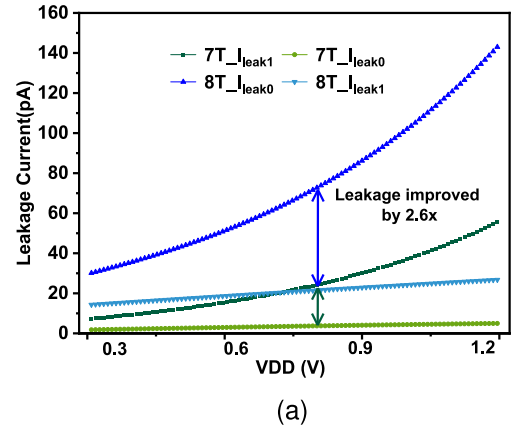


Fig. 9. RBL leakage of the proposed 7T SRAM cell.

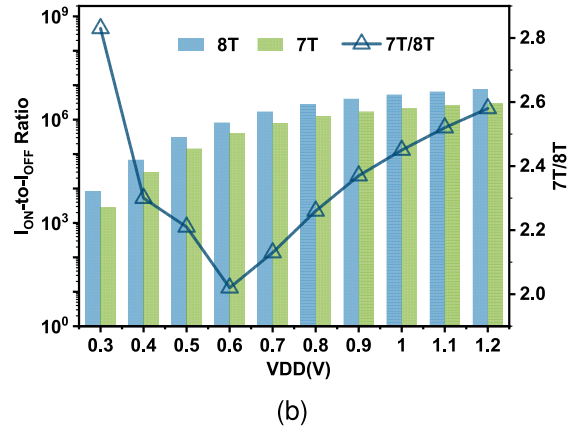
operation waveforms are illustrated in Fig. 8, where RBLA and RBLB represent the RBL waveforms for unselected cells with QBs all store “1” and all store “0,” respectively. In the worst-case scenario, the rising edge of the RBL waveform is delayed by Δt , and the read power consumption increases, as indicated by the shaded region. Therefore, the proposed 7T SRAM with shared charging transistors is better suited for applications with more 0 in the data.

B. Reduced Data-Dependent RBL Leakage for Enhanced Read Sensing Margin

RBL leakage is a major factor affecting the read sensing margin and limits SRAM voltage scaling [27]. As explained in Fig. 1, the 2T RP suffers from data-dependent RBL leakage and a significantly degraded sensing margin at low voltages in the conventional 8T SRAM cell. The proposed 7T SRAM cell uses one pMOS transistor as the RP to improve read operation and lower V_{min} . Fig. 9 depicts the leakage path of the proposed 1T RP. The main leakage path is through the charging pMOS transistor and RP from VDD. Due to the weaker current driving capability of pMOS devices compared to nMOS devices, the RBL leakage in the 7T cell is very low with reduced data dependency. Fig. 10(a) demonstrates the simulated I_{leak0} and I_{leak1} of the 8T and proposed 7T RBLs. It can be seen that the RBL leakage current of the 7T is much lower than that of the 8T, and the 7T cell RBL leakage shows significant reduced data dependency. Compared to the 8T SRAM cell, the average RBL leakage current of



(a)



(b)

Fig. 10. (a) Simulated I_{leak0} and I_{leak1} of the conventional 8T and proposed 7T RBLs. (b) Improved I_{on} -to- I_{off} ratio of proposed 7T.

the 7T SRAM cell is improved by $2.6\times$. The standby power consumption (the average of all cells store “0” and all cells store “1”) in a 64×64 array of 7T SRAM decreases by 22.8% and 23% compared to that of 8T SRAM at 0.5 and 1.2 V, respectively. Fig. 10(b) shows the I_{on} -to- I_{off} ratio of 1T and 2T RPs with respect to VDD. In comparison with conventional 8T, the proposed 7T SRAM improves I_{on} -to- I_{off} ratio by $2.8\times$ and $2.3\times$ at 0.3 and 0.4 V, respectively. Therefore, the 7T cell demonstrates better read stability at low voltages. The benefits of reduced standby power and improved read stability make our SRAM designs well suited for battery-operated and energy-constrained devices.

IV. TIMING GENERATION AND ERROR DETECTION

In the subthreshold region, the transistor I_{on} -to- I_{off} ratio changes significantly with PVT variations. The read current may not be distinguishable from the total RBL leakage current. If the pulse width of RWL is too short, the bitline may not be charged to the switching threshold voltage of the read buffer, resulting in read “1” error; if the pulse width of RWL is too long, the bitline may be charged by the leakage current to reach the switching threshold voltage of the read buffer, resulting in read “0” error. And the active power consumption of RBLs during read “1” operation will also increase. Conventional timing generation circuits typically generate precharge signals

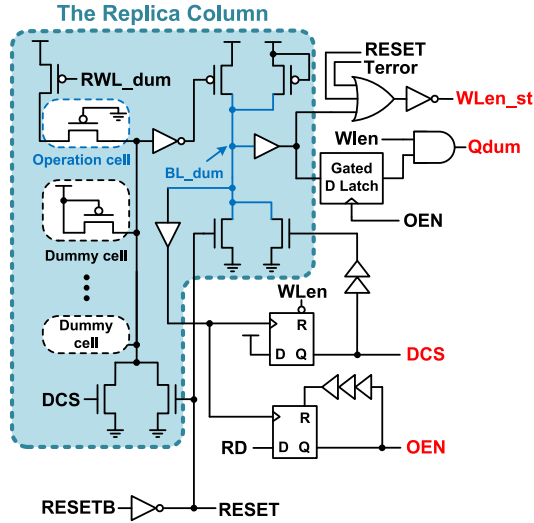


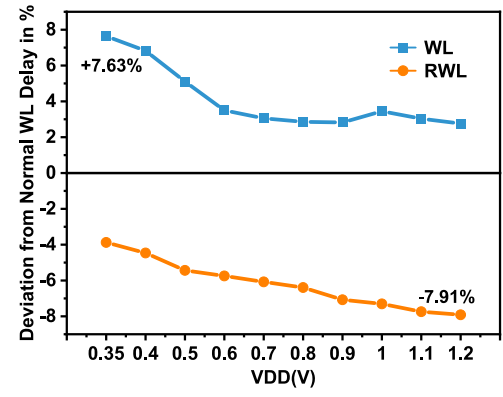
Fig. 11. Schematic of the DTBATG circuits.

through a combinational logic delay chain, which does not provide the flexibility to cover access time deviations caused by PVT variations, thus restricting SRAM voltage scaling. Therefore, we apply a timing generation circuit that can adjust the RWL pulse with by tracking the delay time of the slowest bitline, thereby optimizing timing margins and dynamic power consumption and improving SRAM read and write stability over a wide voltage range.

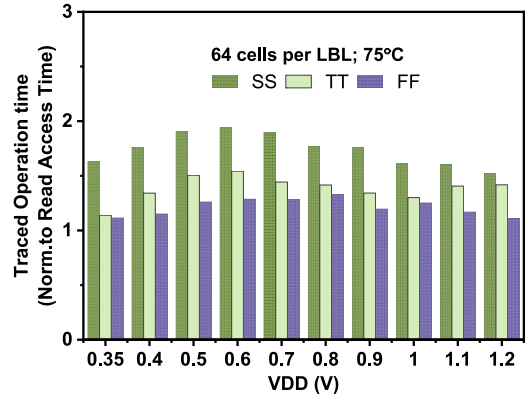
A. Delay-Tracking-Based Adaptive Timing Generation Circuit

Fig. 11 illustrates the detailed structure of the proposed DTBATG circuit. The DTBATG circuit consists of a read “1” replica column and a timing control module. The read “1” replica column consists of dummy cells and an operation cell. RWL is generated from word line (WL). We increase the number of dummy cells on the replica column to generate WL with appropriate pulse width. When the 7T SRAM performs a read/write operation, the WL_EN flag first becomes “0” if $CENB$ is asserted, and then the WL corresponding to the address becomes “1,” followed by the replica column read “1” operation with the control signals WL_dum and RWL_dum . These two signals are generated by a replica path that mimics the address decoder delay of regular WL and RWL signals. This ensures that the operation cell on the replica column is simultaneously activated as the regular SRAM cells in the array when a read/write operation is performed. When the local bitline of the replica column is charged to the switching threshold voltage of the read buffer, the global bitline BL_dum of the replica column becomes “1” and then the WL_EN flag is set to “1,” which negate WL to “0.” At the same time, DCS is asserted to perform bitline refreshing. During the read operation cycle, after the replica column global bitline is charged to “1,” a narrow pulse OEN will be generated to enable the output latch.

The DTBATG circuit adjusts the WL pulse width by tracking the delay time of the slowest bitline. As the read “1” operation reflects the worst-case bitline delay and the write



(a)



(b)

Fig. 12. (a) Delay deviation between the replica path and the normal path. (b) Traced operation time normalized to RBL delay time.

delay is less than the read delay, the WL width generated by DTBATG can simultaneously cover the maximum access time required by both the write and read operations. Furthermore, since the delay time of bitline discharge in the replica column is affected by PVT variations, DTBATG can track the PVT variations to adaptively adjust the WL pulse width. This adaptive adjustment optimizes dynamic power consumption by controlling the bitline charge time while simultaneously improving SRAM read and write stability to PVT variations. Fig. 12(a) depicts the comparative delay results between the replica column and the normal path under various voltages. Post-layout simulations were conducted at different temperatures (-10°C , 25°C , and 75°C) and process corners (slow-slow [SS], typical-typical [TT], and fast-fast [FF]), with the worst-case results chosen for display at each voltage group. The results indicate that the delay deviation of the replica column relative to the normal path remains within 8% with PVT variations. Furthermore, RWL_dum exhibits a slight delay compared to the normal RWL. This variation is acceptable and will not lead to read failures. Fig. 12(b) shows the traced operation time normalized to RBL delay time at different process corners with temperature of 75°C . The DTBATG circuit demonstrates good tracking effect from 0.35 to 1.2 V. The worst case occurs at FF corner, 1.2 V, with timing margin of 12%.

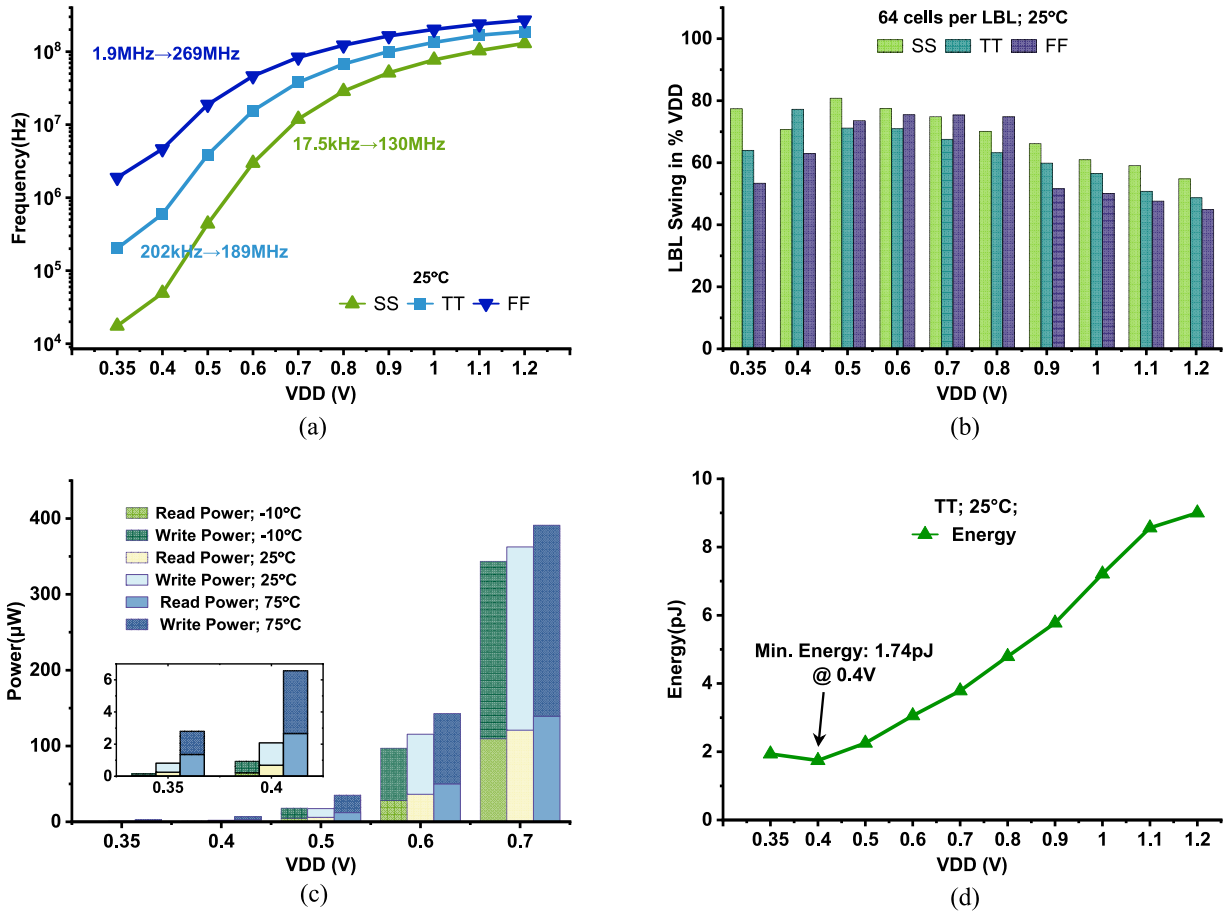


Fig. 13. Simulated results of the proposed 7T SRAM: (a) operating frequency, (b) LBL swing, (c) write and read power, and (d) energy.

B. Error Detection

Based on the DTBATG circuit, we propose a lightweight read error detection circuit to ensure the correct operation of the SRAM during supply voltage scaling. As shown in Fig. 6, the error detection circuit includes another enable signal, Rpo_n , in addition to the data output $Qdum$ of the replica column. Once the Rpo_n is asserted, if the $Qdum$ does not reach “1” at the rising edge of the clock comes after the read operation cycle, the error flag (*Error*) is triggered and sent to the control system. When a read error occurs, the output enable signal, OEN, is disabled until the error flag is reset. Errors in the SRAM read operations can be corrected by reducing the frequency or raising the supply voltage and replaying the instruction. The DTBATG-based error detection circuit avoids complex mechanisms and contributes negligible area and energy overhead to the SRAM macro.

V. MEASUREMENT AND ANALYSIS

A 128-row, 64-column SRAM macro with a total capacity of 8 kb is implemented in HHGrace 55-nm standard performance (SP) CMOS process, with four SRAM cells in each row sharing a charging pMOS. The 8-kb SRAM is post-simulated using cadence virtuoso tool at different process corners (SS, TT, and FF) and temperatures (−10 °C, 25 °C, and 75 °C),

and the implemented 8-kb SRAM measurements is performed in a voltage-scalable test system on chip (SoC).

A. Post-Layout Simulation Results

The post-layout simulation results show that the proposed SRAM is able to achieve full read and write functionality with a minimum voltage of 0.35 V at all process corners and temperatures. The clock period is set to $1.2 \times$ of the maximum read access time. Fig. 13(a) shows the operating frequency of the proposed 7T SRAM with respect to VDD over a range of 0.35–1.2 V in the step of 0.1 V at different corners. At TT corner, 25 °C, the corresponding SRAM operating frequencies are 202 kHz and 189 MHz at 0.35 and 1.2 V, respectively. The SRAM access speed increases slowly around 1 V, which is caused by the timing margin introduced in the DTBATG circuit to balance SRAM performance in both the $sub - V_t$ and the $super - V_t$ regions. Fig. 13(b) shows the simulated local bitline swing with respect to VDD at different corners. The LBLs do not reach full swing because the charging is stopped after the LBLs reach the switching threshold voltage of the read buffer to save active power. Fig. 13(c) shows the simulated power of write and read “1” operations at TT corner. The access energy of the proposed SRAM, which is the average energy of four consecutive cycles of read “1,” write “0,” read “0,” and write “1” operations, is shown in Fig. 13(d). The minimum energy consumed is 1.74 pJ at 0.4 V.

TABLE II
PERFORMANCE COMPARISON WITH PRIOR SRAMS

	[15]	[28]	[29]	[30]	[31]	Compiler	This work	
Cell type	9T	8T	8T	6T	13T	6T	7T	
Results	Measured	Measured	Measured	Measured	Measured	Measured	Simulated	Measured
Process	40nm	65nm	55nm	55nm	65nm	55nm	55nm	
Assist Technique	NBL +WLOD	Data Encoding	Adaptive VVSS	WLUD +DAWA	WLOD	MT-MOS	w/o	
Cell size(μm^2)	0.845	1.56	1.17	0.62	12.44	0.425	1.66 ^L	
Capacity(Kb)	72	32	512	128	16	64	8	
Word Length	32	32	32	16	32	32	32	
VDD(V)	0.325	0.36	0.6	0.6	0.3	1.2	0.35	0.75
Frequency@VDD	600kHz	250kHz	209MHz	360MHz	0.5kHz ^E	414MHz	202kHz	32MHz
Active Power($\mu\text{W}/\text{MHz}$)	9.63	1.48	20	23	21.3	8.26	1.94	6.65 ^E

^LLogic Rules. ^EEstimated.

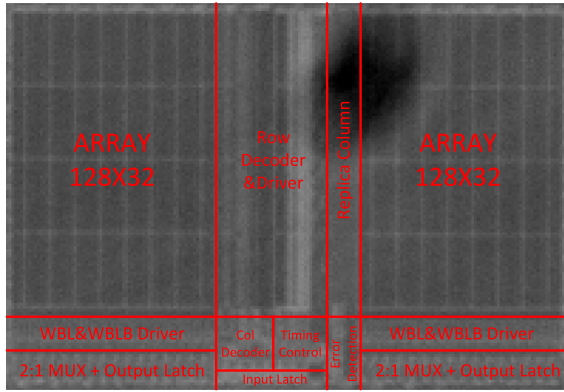


Fig. 14. Die photograph of the implemented 8-kb SRAM.

Table II shows the performance of the proposed 7T SRAM in comparison with other recent works [15], [28], [29], [30], [31] and a standard 6T SRAM generated by memory compiler. At 0.4 V, the proposed 7T SRAM operates at 595.3 kHz, consuming 1.16 and 2.33 pJ for read and write operations, respectively. At 25 °C, the 8-kb SRAM has an average leakage current of 260.7 nA. The active power consumption of the proposed 7T SRAM is 1.94 and 4.35 $\mu\text{W}/\text{MHz}$ at 0.35 and 0.75 V, respectively. In comparison with other listed works in CMOS process, our design shows improved energy efficiency, and operates over a wide frequency range (202 kHz–189 MHz) for better dynamic voltage scalability (0.35–1.2 V). This feature makes our design compatible with many high reliability and energy efficiency applications.

B. Test Chip Implementation and Analysis

The die photo of the implemented 8-kb SRAM with module locations is shown in Fig. 14. The proposed 8-kb 7T SRAM and a 64-kb standard 6T SRAM generated by memory compiler are co-located on a voltage-scalable SoC as optional on-chip memory components. The maximum operating frequency of the test chip is 32 MHz. The standard 6T SRAM is applied with HHGrace 55-nm low-power design rule and implemented with a high-density layout structure,

while the 7T SRAM layout is implemented under logic rules, which results in larger area and power consumption than 6T SRAM of the same size. The 7T SRAM is capable of achieving full write functionality and partial read functionality within the voltage range of 0.75–1.2 V. In the worst-case scenario where all unselected QBs in the same column are store “0,” attempting to read 32 bits of “1” simultaneously may result in errors. This occurs because the charging pMOS transistors in each column are designed with limited drive strength, and the replica column do not adequately reflect the worst-case conditions of the read operation. The actual minimum operating voltage of the implemented SRAM cannot be accurately measured. This is because, even at the lowest operating frequency (1 MHz), the processor fails to function properly when the supply voltage drops below 0.75 V, resulting in the malfunction of the voltage scaling control algorithm. In the best-case scenario where all unselected QBs in the same column are store “1,” the average power consumption of 7T SRAM for read and write operations (write “0,” read “0,” write “1,” and read “1”) is measured. At 0.75 V, 32 MHz, the average power consumption of the entire chip performing 7T SRAM read and write operations is 9.75 mW and is 19.3% lower compared to performing standard 6T read and write operations, which consumes an average power of 12.1 mW. The measurement results exhibit a power reduction trend similar to that of the simulation results, indicating that the 7T SRAM shows power advantages under low-voltage conditions but may be influenced by the distribution of QBs in specific scenarios, warranting careful consideration and better timing design.

VI. CONCLUSION

A low-power subthreshold 7T SRAM with enhanced bitline leakage and read sensing margin to improve $V_{\min}$ has been demonstrated in this article. The bitline leakage current for 7T SRAM cell is improved by 2.6 $\times$ and the $I_{\text{on-to-}I_{\text{off}}}$ ratio is improved by 2.3 $\times$ compared to conventional 8T SRAM cell. We employ a novel read scheme without precharge operation to lower dynamic power dissipation during read operations. Furthermore, a DTBATG circuit and an error detection circuit

are implemented in the design for PVT variation-tolerant operations and reliable dynamic voltage scaling. An 8-kb SRAM has been implemented in 55-nm CMOS technology. The measurement results indicate that 7T SRAM holds potential for low-voltage operation and is expected to be used in power-sensitive products. However, further research is required to enhance its reliability and performance.

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